

# UDAY OJHA

Chandigarh, India | +91 8360493330 | udayojha129@gmail.com  
oj-uday.github.io | github.com/OJ-Uday | linkedin.com/in/uday-o

## Summary

Software engineer (B.Tech CS '26). At Eli Lilly, designed and shipped a production telemetry platform end-to-end — TypeScript browser SDK, Go + ClickHouse analytics backend, React dashboard, zero-touch Kubernetes/GitOps delivery. Independently authored vetlock, an open-source npm supply-chain scanner (17 detectors, 375 tests, 12/13 historical attacks caught, 31 red-team exploits closed) with a live browser demo on Cloudflare Workers + GitHub Actions. CTF reverse-engineering lead; two prior cybersecurity internships.

## Technical Skills

**Languages:** TypeScript, JavaScript, Go (Golang), Python, Java, C, C#, SQL, Bash

**Backend & Data:** Node.js, REST APIs, microservices, event pipelines, ClickHouse, MySQL, PL/SQL

**Frontend:** React, micro-frontends (Module Federation), Jest, React Testing Library, HTML, CSS

**Cloud & DevOps:** AWS, Kubernetes, Docker, GitOps (Argo CD), GitHub Actions, Cloudflare Workers, CI/CD, Git, npm

**Security:** SAST, supply-chain security, static analysis, AST parsing (Babel), lockfile analysis, Ghidra, IDA Pro, GDB, x86/ARM Assembly, Wazuh (SIEM), Burp Suite, Nmap, Wireshark, penetration testing

## Experience

**Software Engineering Intern** — Eli Lilly and Company — Bangalore, India (On-site) **Jan 2026 – Jul 2026**

- Designed and shipped an end-to-end product-telemetry platform — TypeScript browser SDK, Go + ClickHouse backend, React dashboard — adopted across a federated micro-frontend estate of 5+ internal applications.
- Invented a two-tier UI element-identity scheme and capture-time causality protocol powering an inference engine that auto-names UI elements for one-click approval; new apps onboard with two HTML attributes, zero code changes.
- Built a release-audit service in Go on ClickHouse (append-only deploy ledger + before/after behavior diffs), backtested over 1,648 deploys / 90 days; its first live run flagged a same-day API regression (53% error rate) with session-level evidence.
- Automated develop→Kubernetes delivery with GitHub Actions + GitOps (CI-authored image bumps, programmatic approval, merge queue), cutting deploy lead time from ~1 day of manual coordination to under 15 minutes with zero human steps.
- Root-caused a React render loop that was OOM-killing CI runners — took a 962-test suite from 60+ min to 35 s; published the SDK to a private npm registry with semantic versioning and property-based tests.
- Worked in an AI-augmented workflow (LLM pair-programming, multi-agent orchestration), landing 25+ production pull requests across SDK, backend, and dashboard in the final two weeks.

**Cybersecurity Intern** — CyberRange (Remote)

**Jul 2024 – Nov 2024**

- Tested and hardened penetration-testing challenges for a security-training platform; performed reconnaissance, mobile-security, and vulnerability analysis across the challenge pipeline.

**Cybersecurity Intern** — PGI-Data (Jakarta, Indonesia)

**May 2024 – Jul 2024**

- Deployed Wazuh SIEM on production servers for threat detection; implemented ZTNA (Zero Trust Network Access), VPN, and PAM (Privileged Access Management) solutions.

**Head of Reverse Engineering** — Cryptonite, MIT Manipal CTF Team

**Nov 2022 – Present**

- Lead the reverse-engineering division — training, research, and strategy across x86/ARM binary analysis and vulnerability research — competing in national and international CTFs.

## Projects

- vetlock** (TypeScript / Node, github.com/OJ-Uday/vetlock, Apache-2.0) — open-source npm supply-chain scanner: behavioral diff between two lockfiles, static analysis over extracted tarballs, 17 detectors (install-script injection, env-token harvest, exfil endpoints, obfuscation jumps, typosquat detection, GHSA correlation). Ships CLI, GitHub Action (SARIF), and a live in-browser demo backed by Cloudflare Workers + GitHub Actions. 375 tests; caught 12 of 13 real historical npm attacks; closed 31 of 35 exploits from a six-agent adversarial red-team review. NEVER-EXECUTE canary invariant.
- telemetry-sdk-patterns** (TypeScript, github.com/OJ-Uday/telemetry-sdk-patterns) — zero-dependency browser-telemetry SDK: batching queue with retry/beacon drain, micro-frontend event bridge, pluggable MSAL-style auth adapter, delegated auto-capture; 24 tests, matrix CI, live demo.
- ratelimit** (Go, github.com/OJ-Uday/ratelimit) — goroutine-safe token-bucket and sliding-window rate limiters: stdlib-only, zero background goroutines, injectable clocks, net/http middleware; 0 allocs/op on the hot path.
- react-hooks-kit** (React + TypeScript, github.com/OJ-Uday/react-hooks-kit) — five strictly-typed hooks: cross-tab localStorage sync, churn-free event listeners, undo/redo with history forking, debounced values; 13 tests.
- NiteCTF Challenge Author** (Reverse Engineering) — authored challenges for an international CTF: .NET + SQLCipher database forensics, a PIE binary requiring Z3 SMT solving, and a custom AES-XTS cipher in Go.

## Education

**B.Tech, Computer Science and Engineering** — Manipal Institute of Technology

**2022 – 2026**

CGPA 7.81/10 · Coursework: Algorithms, Data Structures, Operating Systems, Computer Networks, OOP